

Attack-Free Early Warning and Timely PGD Escalation for Catastrophic Overfitting

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Abstract—Single-step adversarial training is efficient but can abruptly lose robustness through catastrophic overfitting (CO). We introduce an attack-free online warning score derived from the cosine consistency of two input gradients already produced by an FGSM parameter update. A threshold selected on three development seeds is frozen before evaluation. It detects 7/7 held-out CIFAR-10/PreActResNet-18 events, 4/5 CIFAR-10/ResNet-18 events, and 2/2 observed held-out CIFAR-100 events, with no alarms in the corresponding non-event controls. Eight reasonable random-start runs at $\epsilon = 8/255$ remain stable and produce no alarms. Under the harder $\epsilon = 16/255$ random-start stress test, all three runs naturally collapse; the frozen detector warns 20–60 updates beforehand in 3/3 cases. Immediate PGD-2 escalation starts before every collapse and raises terminal PGD-10 accuracy from 0 to $26.48 \pm 0.83\%$. A five-seed timing audit shows why immediate action matters: replaying the affected epoch starts PGD-2 before collapse in 5/5 cases, whereas waiting for the next epoch succeeds in only 3/5. Under one training and evaluation harness, the immediate policy reaches $46.83 \pm 0.55\%$ PGD-10 in 18.68 minutes, compared with $47.09 \pm 0.71\%$ in 27.27 minutes for FastAdv+ and $48.31 \pm 0.81\%$ in 45.71 minutes for GradAlign. These results support the monitor as a low-cost mechanism for detecting and acting on unsafe fast training trajectories.

I. INTRODUCTION

Deep classifiers can change their predictions under small adversarial perturbations [1], [2]. Adversarial training addresses this failure by optimizing against perturbed examples and remains a strong empirical defense [3]. Its main obstacle is cost because multi-step training attacks require several forward-backward passes for nearly every parameter update.

Fast adversarial training replaces the inner loop with a single FGSM step. With randomized initialization and a suitable schedule, it can approach the robustness of PGD training at much lower cost [4]. Yet single-step training can exhibit *catastrophic overfitting* (CO): iterative-attack accuracy falls sharply toward zero while clean or FGSM accuracy remains high. FastAdv+ responds using periodic PGD validation [5]; GradAlign regularizes local input-gradient geometry at every update [6]. These approaches improve reliability but add attack or gradient computation.

We ask whether the ordinary FGSM update itself provides an early warning. It computes an input gradient at the attack starting point and an endpoint loss gradient during parameter backpropagation. Retaining the endpoint input gradient exposes their cosine similarity without another model evaluation. A local drop in this similarity becomes our

warning score. When it crosses a frozen threshold, training immediately replays the affected epoch with two-step PGD and continues with PGD-2.

The contributions are:

- An attack-free CO monitor whose threshold is selected on three development seeds and evaluated on held-out seeds, a second residual architecture, CIFAR-100, and a larger threat budget.
- An event-level timing study that records warning, unprotected CO, and actual PGD-2 start times. Immediate escalation acts before collapse in 5/5 matched events, compared with 3/5 for next-epoch escalation.
- Five-seed component controls and direct FastAdv+/GradAlign baselines under the same code path and evaluator, with paired bootstrap confidence intervals for small performance differences.
- A random-start stress test showing that the detector stays silent on eight stable $\epsilon = 8/255$ runs but forecasts 3/3 naturally occurring CO events at $\epsilon = 16/255$, where immediate intervention restores non-trivial robustness.

The study concerns empirical ℓ_∞ robustness. Certified defenses provide formal guarantees through different mechanisms and are complementary to the training-time monitor studied here.

II. RELATED WORK

A. Adversarial Attacks and Training

FGSM follows the sign of the input loss gradient in one step [2]. PGD iterates gradient steps with projection onto a prescribed threat set and is central to the robust-optimization view of adversarial training [3]. TRADES separates natural error from a boundary-related robustness term and makes the clean–robust trade-off explicit [7]. Although these methods are effective, their multi-step inner loops increase training cost. Free adversarial training reuses gradients across repeated mini-batches [8], whereas fast training uses a single randomized FGSM step [4].

Robustness evaluation itself must be sufficiently strong. PGD results can be sensitive to iterations, restarts, and objectives. AutoAttack combines complementary parameter-free attacks to reduce evaluator bias and expose gradient masking [9]. In the controlled study below, we use progressively stronger full-test PGD variants and multiple restarts rather than treating FGSM accuracy as a robustness estimate.

B. Catastrophic Overfitting

Wong *et al.* identified CO as a failure mode of fast adversarial training [4]. Andriushchenko and Flammarion connected it to local nonlinearity and proposed GradAlign, which regularizes gradient alignment inside the perturbation set [6]. Li *et al.* showed that recovery from weak-attack overfitting is important and proposed FastAdv+ and FastAdvW, which use periodic PGD validation to decide when to introduce PGD updates [5]. Our response mechanism is related in spirit, but the detector differs operationally: it consumes the two input gradients already present in the ordinary FGSM parameter update and does not execute a validation attack. We also separate prediction from mitigation by freezing the detector before held-out evaluation.

C. Empirical and Certified Robustness

Empirical defenses are evaluated against known attacks but do not guarantee correctness for every allowed perturbation. Certified methods instead compute a provable lower bound, for example through convex outer approximations [10] or randomized smoothing under an ℓ_2 threat model [11]. This paper targets efficient empirical training in the ℓ_∞ setting. The distinction is important: the attack results reported below measure resistance to a strong test suite, not certified radii.

III. METHOD

A. Threat Model and Fast Update

Let f_θ be a classifier and ℓ cross-entropy loss. Adversarial training under an ℓ_∞ radius ϵ approximates

$$\min_{\theta} \mathbb{E}_{(x,y)} \left[\max_{\|\delta\|_\infty \leq \epsilon} \ell(f_\theta(x + \delta), y) \right]. \quad (1)$$

FGSM begins at x^0 , either x or a random point in the perturbation set, and constructs

$$g^0 = \nabla_{x^0} \ell(f_\theta(x^0), y), \quad x^{\text{adv}} = \Pi(x^0 + \alpha \text{sign}(g^0)), \quad (2)$$

where Π projects onto the valid image and threat set.

B. Attack-Free Gradient-Consistency Score

The endpoint backward pass produces $g^1 = \nabla_{x^{\text{adv}}} \ell(f_\theta(x^{\text{adv}}), y)$ while computing the parameter gradient. We retain it and calculate

$$c(x) = \frac{\langle g^0, g^1 \rangle}{\|g^0\|_2 \|g^1\|_2}. \quad (3)$$

Batch averages are accumulated over non-overlapping windows of 20 optimizer updates to give c_t . The monitor adds reductions and trace state, but no forward or backward pass. To remove the slow training trend, it uses

$$b_t = \text{median}(c_{t-5}, \dots, c_{t-1}), \quad s_t = b_t - c_t, \quad (4)$$

after at least three history windows. A warning occurs when $s_t \geq \tau$. Seeds 17, 23, and 42 select $\tau = 0.1663298875$ for CIFAR-10; the rule is then frozen. CIFAR-100 uses three development seeds to calibrate its score scale once, followed by a disjoint held-out evaluation.

C. Immediate Conditional Escalation

The primary response snapshots model, optimizer, mixed-precision scaler, loader, and random states at each epoch boundary. At the first warning it stops further FGSM updates, restores the epoch-start state, and immediately replays that epoch using PGD-2. All later epochs also use PGD-2. This makes the first PGD attack begin with zero additional FGSM updates after the warning.

We compare this response with three matched alternatives: retain the current FGSM epoch and start PGD-2 at the next epoch, switch at a fixed early or late epoch, and use PGD-2 from epoch 1. The replay is called “rollback” in the tables. Its purpose is temporal: the accuracy ablation alone shows little mean gain, whereas the event audit tests whether PGD-2 actually begins before the unprotected collapse.

D. Event and Forecast Protocol

On fixed validation data, a CO event requires PGD-10 accuracy to fall at least 20 percentage points from its previous peak and reach at most 10%. We report whether a warning precedes the first event, whether it occurs within 60 updates, median lead, and alarm episodes in non-event runs. Validation PGD is used only for retrospective event labels; it never triggers the proposed monitor. Recall intervals are Wilson 95% intervals. Accuracy and time comparisons use identical seeds and paired bootstrap 95% intervals with 100,000 resamples.

IV. EXPERIMENTS

A. Setup

The primary setting is CIFAR-10 with PreActResNet-18 [12], [13], a deterministic 45,000/5,000 training/validation split, and $\epsilon = 8/255$. All methods use SGD with momentum 0.9, weight decay 5×10^{-4} , batch size 128, mixed precision, 30 epochs, and a triangular learning-rate schedule. The collapse-prone recipe uses zero-start FGSM, step 8/255, and peak learning rate 0.3. The ordinary FGSM-RS control uses a uniform random start, step 10/255, and peak learning rate 0.2. Recovery uses PGD-2 with step 4/255.

Transfer tests change one factor at a time: torchvision ResNet-18 on CIFAR-10, and PreActResNet-18 on CIFAR-100. CIFAR-100 uses three development seeds and held-out seeds 101, 202, and 303 plus two held-out random-start controls. A separate CIFAR-10 stress test retains random starts and uses $\epsilon = 16/255$, FGSM step 20/255, and peak learning rate 0.2.

B. Detection and Transfer

The original held-out CIFAR-10 evaluation contains seven events spanning epochs 13–24. The frozen monitor warns within 60 updates in 7/7 cases, with 40 updates median lead and one earlier alarm episode. It emits no alarm on three ordinary FGSM-RS runs. On the same traces, a periodic PGD-10 probe every 100 updates detects 6/7 events with a median delay of 40 updates after the event and makes a median of 73 PGD calls before failure.

Table I extends the evaluation. The unchanged CIFAR-10 threshold detects 4/5 ResNet-18 events. For CIFAR-100, three development seeds calibrate a domain-specific threshold; two of three planned held-out collapse-prone runs exhibit the registered event and both are detected. No alarm occurs in either CIFAR-100 random-start control. Wilson intervals are wide because event counts are small, but the results establish transfer beyond one architecture and one dataset without presenting aggregate percentages as certainty.

C. Natural CO under Randomized FGSM

The detector should not require a deliberately weak zero-start attack. We therefore first test eight randomized FGSM runs at $\epsilon = 8/255$: three standard runs at peak learning rate 0.2 and five held-out runs at 0.3. All eight remain robust, and the detector produces zero alarm episodes. Their stability confirms that random starts are effective at this budget and that the monitor does not add unnecessary PGD training.

Prior work reports that random starts become less reliable at larger budgets [6]. At $\epsilon = 16/255$, all three held-out random-start runs naturally meet the CO definition and terminate at 0% PGD-10. The frozen detector warns 20, 40, and 60 updates before collapse. Immediate PGD-2 begins before every event and finishes at 25.55%, 26.80%, and 27.11% PGD-10, respectively. Mean clean and PGD-10 accuracy are $71.05 \pm 0.43\%$ and $26.48 \pm 0.83\%$. This stress result demonstrates value on naturally occurring randomized-FGSM collapse rather than only on the zero-start construction; its accuracy is reported separately from the 8/255 benchmark.

D. Actionable Lead and Component Controls

Table II distinguishes statistical lead from usable lead. Both responses receive the identical first warning. Immediate epoch replay begins PGD-2 with no intervening FGSM update and acts before all five matched collapses. Waiting for the epoch boundary adds a median of 212 FGSM updates; two collapses occur before PGD-2 starts. Final accuracy differs little, so the reason to retain immediate replay is timely action rather than a claimed accuracy boost.

All component controls now use the same five seeds. As shown in Table III, immediate replay and next-epoch escalation differ by only +0.11 PGD-10 points (paired bootstrap 95% CI $[-0.52, +0.77]$) while replay adds 0.53 min. Full PGD-2 differs from the next-epoch policy by +0.09 points (CI $[-0.27, +0.42]$) but adds 3.46 min. The detector therefore selects when to incur PGD computation; PGD-2 itself provides the recovery.

E. Direct CO-Mitigation Baselines

FastAdv+ and GradAlign are rerun with the same data split, architecture, seeds, 30-epoch schedule, checkpoint rule, and PGD-10 evaluator. Table IV gives a direct numerical comparison. FastAdv+ is 0.38 points higher than the next-epoch policy, but the paired CI $[-0.34, +1.11]$ includes zero; it takes 9.12 additional minutes (CI $[5.86, 11.88]$).

GradAlign gains 1.59 PGD-10 points and costs 27.57 additional minutes. These results locate the proposed policy on the robustness-cost curve: its primary advantage is attack-free online evidence and conditional computation.

F. Strong Test-Set Evaluation

Representative frozen checkpoints are evaluated on all 10,000 CIFAR-10 test images. PGD-50 and five-restart PGD-20 agree within 0.2 points, providing no evidence of a weak iterative evaluation. Immediate replay retains 45.46% PGD-50; ordinary FGSM-RS reaches 46.24%, and 110-epoch PGD-10 training reaches 50.77% at substantially higher training cost.

V. DISCUSSION AND REPRODUCIBILITY

The experiments separate four roles. The cosine drop supplies attack-free evidence; PGD-2 restores robustness; immediate replay turns warning lead into actionable lead; and rollback has no reliable final-accuracy advantage by itself. This separation avoids attributing the full improvement to the detector when the stronger training attack performs the recovery.

Table VI exposes every matched timing event instead of only the aggregate rate. The five trajectories cover three operational regimes. A warning can coincide with an epoch boundary (seed 17), occur far before CO (seeds 23 and 202), or leave only 20–60 updates (seeds 42 and 101). An epoch-boundary response is adequate in the first two regimes but fails in both short-lead cases. Immediate replay has zero warning-to-start delay in all five cases because it restores the saved epoch-start state and reprocesses the affected epoch with PGD-2. Thus, the benefit of replay is not a higher mean endpoint; it is a deterministic upper bound of zero additional unsafe FGSM updates after the first alarm.

Architecture transfer is imperfect: unchanged-threshold recall falls from 7/7 on PreActResNet-18 to 4/5 on ResNet-18, although no control alarm occurs. CIFAR-100 requires one development calibration because score scale changes, then detects both observed held-out events. The Wilson lower bounds in Fig. 1 reflect the small event samples. The evidence supports transfer of the signal and protocol, while leaving room for larger architecture and dataset studies.

Every detector choice is fixed before its associated held-out evaluation: 20-update windows, five-window median history, three-window minimum history, the event rule, and the 60-update near horizon. Fixed-switch controls ignore detector callbacks. Training time includes core parameter updates and excludes retrospective label attacks for every method. The released audit tables retain seed-level warning, collapse, and PGD-start steps, and all confidence intervals are generated from the saved per-seed results.

The online storage and compute path is deliberately small. During the normal FGSM update, the implementation retains the endpoint input gradient only until its cosine with the attack-start gradient is reduced to a batch scalar. The monitor then stores window summaries rather than images or gradient tensors. No validation attack is called by the decision

TABLE I

DETECTOR TRANSFER. “NEAR” MEANS A WARNING WITHIN 60 UPDATES; CONTROLS COUNT NON-EVENT RUNS AND EVERY ALARM EPISODE; LEAD IS IN UPDATES.

Dataset / budget	Architecture	Planned	Events	Detected	Near	Controls / alarms	Median lead
CIFAR-10, 8/255	PreActResNet-18	7	7	7/7	7/7	3 / 0	40
CIFAR-10, 8/255	ResNet-18	5	5	4/5	2/5	3 / 0	80
CIFAR-100, 8/255	PreActResNet-18	3	2	2/2	2/2	2 / 0	40
CIFAR-10, 16/255 RS	PreActResNet-18	3	3	3/3	3/3	–	40

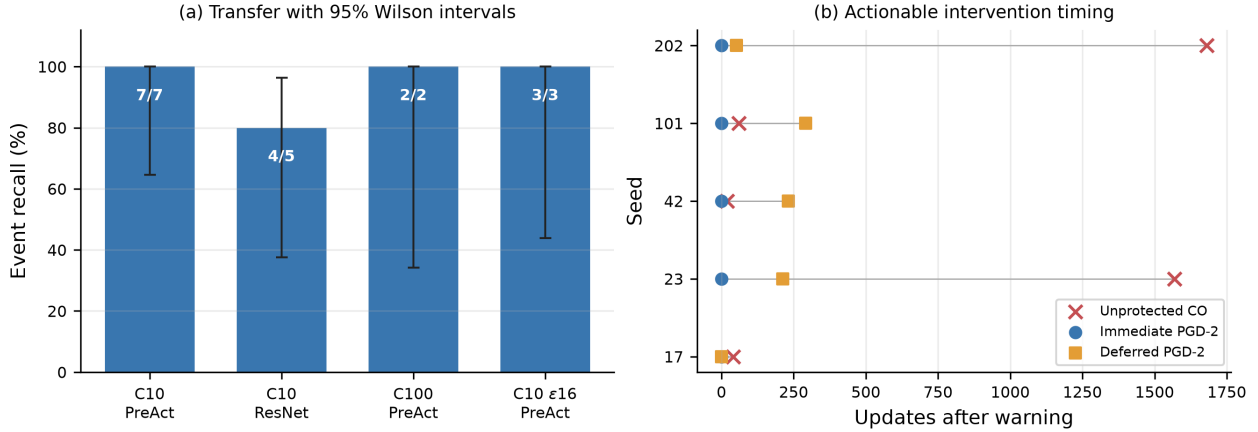


Fig. 1. Left: event recall across architecture, dataset, and threat-budget transfer, with Wilson 95% intervals. Right: after a warning, immediate epoch replay starts PGD-2 at once; a next-epoch switch starts after the unprotected CO in two of five matched trajectories.

TABLE II
FIVE-SEED WARNING-TO-ACTION AUDIT

Response	Before CO	Warn→start	PGD-10
Immediate epoch replay	5/5	0 updates	46.83 ± 0.55
Next epoch	3/5	212 updates	46.72 ± 1.05

TABLE III
MATCHED FIVE-SEED CONTROLS (MEAN ± SAMPLE STD.)

Policy	Clean	PGD-10	Minutes
Immediate replay	83.14 ± 0.62	46.83 ± 0.55	18.68 ± 0.70
Next epoch	83.09 ± 0.57	46.72 ± 1.05	18.14 ± 0.58
Fixed epoch 13	83.32 ± 0.70	46.41 ± 0.50	19.37 ± 0.21
Fixed epoch 21	82.35 ± 1.12	45.75 ± 0.94	17.85 ± 0.21
PGD-2 from epoch 1	83.57 ± 1.02	46.81 ± 0.90	21.60 ± 0.13

TABLE IV
UNIFIED FIVE-SEED CO-MITIGATION BASELINES

Method	Clean	PGD-10	Minutes
Ours, next epoch	83.09 ± 0.57	46.72 ± 1.05	18.14 ± 0.58
Ours, immediate	83.14 ± 0.62	46.83 ± 0.55	18.68 ± 0.70
FastAdv+	81.96 ± 0.60	47.09 ± 0.71	27.27 ± 3.53
GradAlign	79.54 ± 0.86	48.31 ± 0.81	45.71 ± 0.11

TABLE V
FULL-TEST CIFAR-10 ACCURACY (%)

Method	Clean	PGD-10	PGD-50	PGD×5
Immediate replay	82.97	47.20	45.46	45.65
Next epoch	82.57	46.38	44.75	44.91
Fast FGSM-RS	82.48	47.77	46.24	46.36
PGD-10 training	82.47	51.76	50.77	50.84

rule. Checkpointing at epoch boundaries is already common practice; replay is invoked once at the first alarm, after which training remains on PGD-2. Consequently, runtime depends on when a trajectory warns instead of paying a multi-step inner-loop cost from epoch 1.

The natural-CO experiment also clarifies the monitor’s intended operating point. At 8/255, randomized initialization prevents collapse in all eight tested runs, and zero alarms leave the efficient recipe untouched. At 16/255, the same broad class of randomized training produces three genuine collapse events. The fixed monitor detects all three, and immediate escalation acts before each event. This pair of regimes is more informative than testing only a hand-

designed zero-start failure: the rule remains inactive when the ordinary recipe is safe and becomes useful when the threat budget makes that recipe unstable.

The controlled comparisons identify a practical trade-off rather than a universal ranking. GradAlign obtains the highest mean PGD-10 accuracy among the 30-epoch fast methods, while FastAdv+ is intermediate in accuracy and time. Conditional escalation uses the least time and supplies an explicit event log linking the signal, decision, and response. Full PGD-2 has nearly identical mean PGD-10 accuracy to conditional escalation but consumes more computation; the full PGD-10 training reference is stronger still under the 110-epoch schedule. Which operating point is preferable

TABLE VI
EVENT-LEVEL TIMING IN OPTIMIZER UPDATES

Seed	Warn→CO	Immediate	Deferred	Deferred before CO
17	40	0	0	yes
23	1568	0	212	yes
42	20	0	232	no
101	60	0	292	no
202	1680	0	52	yes

therefore depends on whether the deployment prioritizes peak robustness, training cost, or auditable failure detection.

Finally, the monitor measures empirical training stability rather than a certified radius. Convex relaxation [10] and randomized smoothing [11] address formal guarantees and can be combined with training-time diagnostics.

VI. CONCLUSION

We presented an attack-free early-warning monitor for catastrophic overfitting and tested whether its warning arrives soon enough to change training. The detector transfers across seeds, ResNet variants, CIFAR-10 and CIFAR-100, and a larger threat budget, while remaining silent in the tested random-start non-event runs. Randomized FGSM at $\epsilon = 16/255$ supplies a natural CO test: all three events are detected, immediate PGD-2 begins before collapse, and terminal PGD-10 rises from zero to $26.48 \pm 0.83\%$. Five-seed controls show that waiting until the next epoch misses the actionable window in two cases, while fixed schedules and full PGD-2 separate detector timing from recovery. Direct FastAdv+ and GradAlign baselines show the trade-off: the proposed policy gives slightly lower robustness with substantially lower training time and no online validation attack. The resulting contribution is a measurable link from free within-update geometry, to early warning, to timely conditional computation.

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